Practical Experiment: Compare the response speed of open-source Large Language Models (LLMs) to find the fastest model for real-time applications

Meeting 5 Packaging for Pip Install

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Version: 1.0

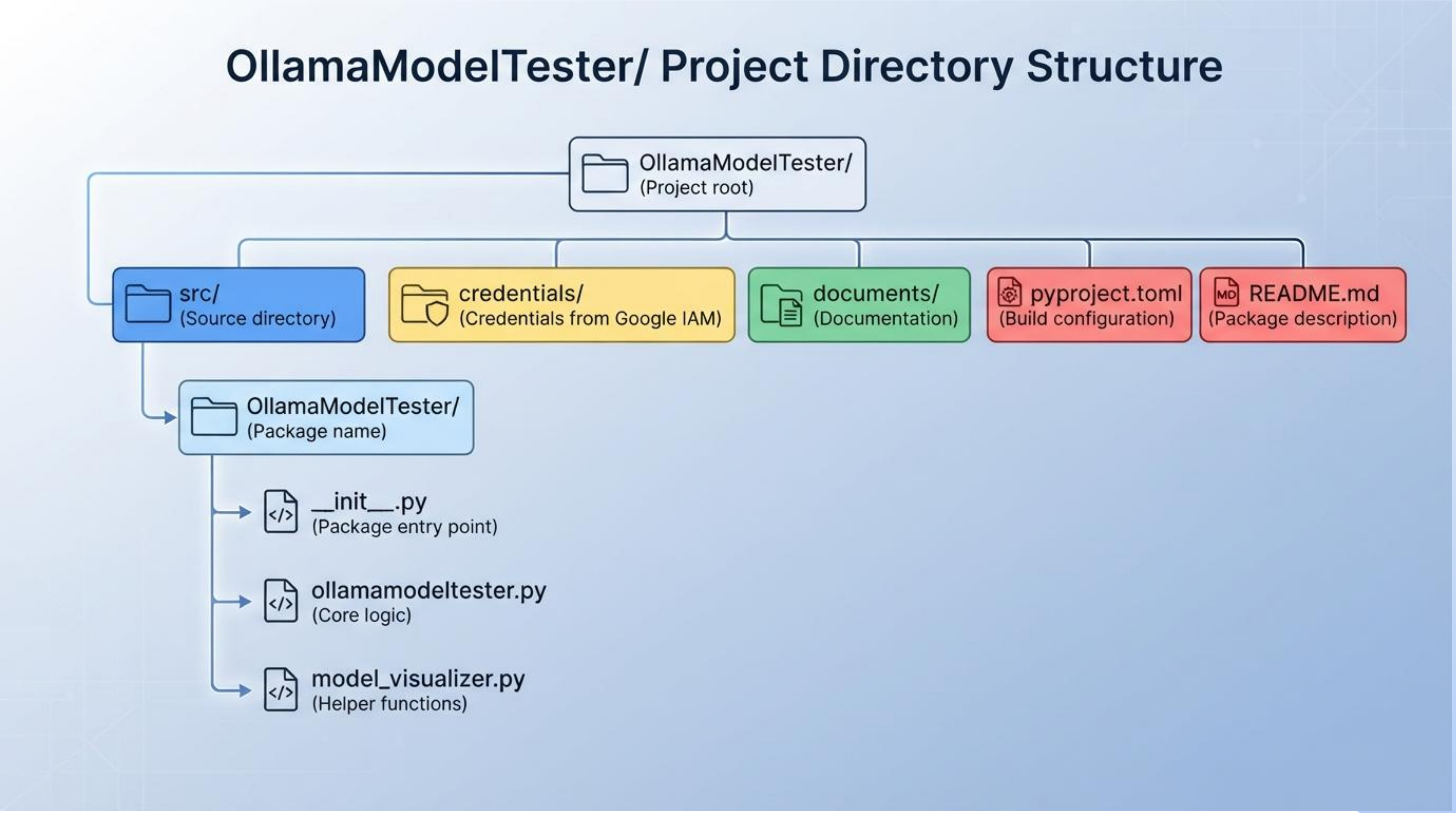
Agenda

- Before PIP vs After PIP
- Package Architecture
- Project Configuration
- Package Structure & Exports
- Build & Packaging Process
- Upload to TestPyPI
- Deployment to PyPI
- Common Issues & Solutions

Before PIP vs After PIP

Before	After
Manual file copying	pip install mypackage
Complex setup	Automatic dependencies
No versioning	Semantic versioning
Local only	Global distribution

Package Architecture



Project Configuration

pyproject.toml Structure

- **Modern** – Fully implements PEP 621 (metadata in pyproject.toml)
- **Simple** – No setup.py or MANIFEST.in needed
- **Reproducible** – Builds are deterministic and reliable

```toml
[build-system]
requires = ["hatchling"]
build-backend = "hatchling.build"

[project]
name = "ollamamodeltester"
version = "0.0.1"
description = "Framework for LLM evaluation"
readme = "README.md"
license = "MIT"
authors = [{name = "Name"}]
requires-python = ">=3.13"
dependencies = [

]
```

Package Structure & Exports

__init__.py Configuration

```python
# src/ollamamodeltester/__init__.py

from .main import OllamaModelTester
from .utils import helper_function

__version__ = "0.0.4"

__all__ = ['OllamaModelTester', 'helper_function']
```

User Writes	Package Needs
from ollamamodeltester import *	__all__ list defines what's imported
from ollamamodeltester import OllamaModelTester	__init__.py must expose it
import ollamamodeltester	Package-level attributes available

Build & Packaging Process

Step 1: Build the Package

```
# Install build tools
pip install build twine
# Build distribution files
python -m build
```

Step 2: Check Package Quality

`$ twine check dist/*`

- Verifies metadata
- Checks for common errors
- Ensures README renders correctly

```
dist/
├── ollamamodeltester-0.0.1-py3-none-any.whl
└── ollamamodeltester-0.0.1.tar.gz
```

.whl : Binary distribution (pre-built) – faster install
.tar.gz : Source distribution – for platforms without wheels

Upload to TestPyPI

Step 1: Create Account

- Go to **test.pypi.org**
- Register and create an **API token** (more secure than password)

Step 2: Upload Package

```
$ twine upload --repository testpypi dist/*
```

Step 3: Authenticate

- **Username**: __token__ (exact string)
- **Password**: pypi- + your API token

Step 4: Test Installation

```
$ pip install -i https://test.pypi.org/simple/ OllamaModelTester
```

Deployment to PyPI

Step 1: Clean folder and final build

```
$ rm -rf dist/
$ python -m build
$ twine check dist/*
```

Step 2: Upload to Live PyPI

```
$ twine upload dist/*
```

Step 3: Verify

Check PyPI project page
https://pypi.org/project/ollamamodeltester/

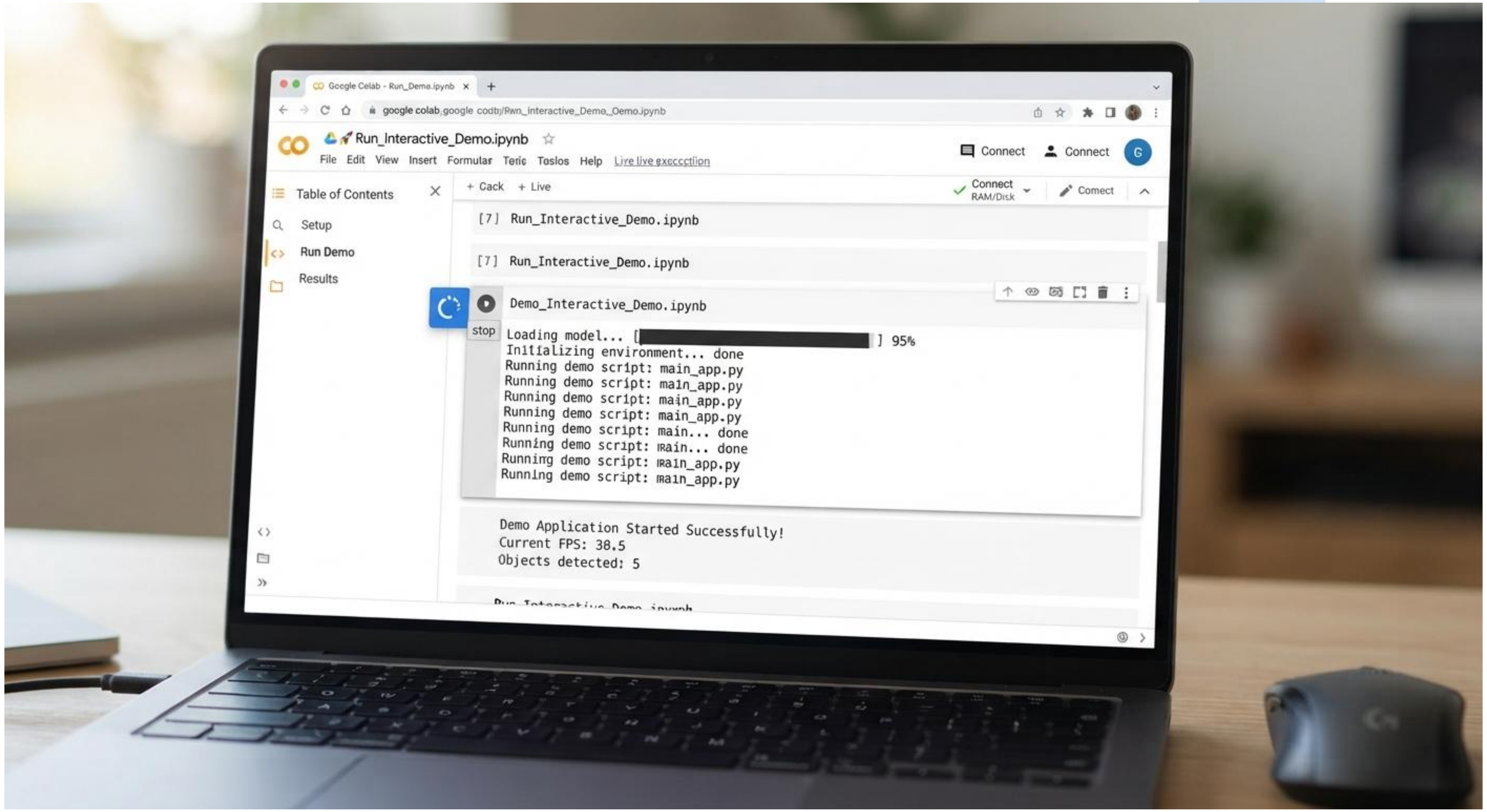
Install from live PyPI

```
$ pip install ollamamodeltester
```

Common Issues & Solutions

Issue	Cause	Solution
ModuleNotFoundError	Package not installed correctly	pip install -e . for development
ImportError on package	Wrong import path in __init__.py	Use relative imports: from .module import Class
twine **not found**	Tool not installed	pip install twine
Upload error (400)	Version already exists	Update version number
Package not searchable	Search index delay (TestPyPI)	Wait 1-2 hours or use direct URL
Dependencies missing	Dependencies not listed in pyproject.toml	Add all required packages to dependencies
README not rendering	Incorrect format or path	Set readme = "README.md" with correct path

LIVE DEMO


Google Colab - Run_Demo.ipynb
Run_Interactive_Demo.ipynb
File Edit View Insert Formular Teric Toslos Help
Connect
Connect
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Setup
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Connect
RAM/Disk
Comect
[7] Run_Interactive_Demo.ipynb
[7] Run_Interactive_Demo.ipynb
Demo_Interactive_Demo.ipynb
stop
Loading model... [ ] 95%
Initializing environment... done
Running demo script: main_app.py
Running demo script: main_app.py
Running demo script: main_app.py
Running demo script: main_app.py
Running demo script: main... done
Running demo script: main... done
Running demo script: main_app.py
Running demo script: main_app.py
Demo Application Started Successfully!
Current FPS: 30.5
Objects detected: 5

References

- Installing Packages – https://packaging.python.org/en/latest/tutorials/installing-packages/
- Packaging Python Projects – https://packaging.python.org/en/latest/tutorials/packaging-projects/
- PyPI – https://pypi.org/
- Test PyPI – https://test.pypi.org/
- GNU General Public License – https://www.gnu.org/licenses/gpl-3.0.html
- Classifiers – https://pypi.org/classifiers/

Thank you

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